

Data Governance

Build trusted, secure, and usable data to power decisions and AI

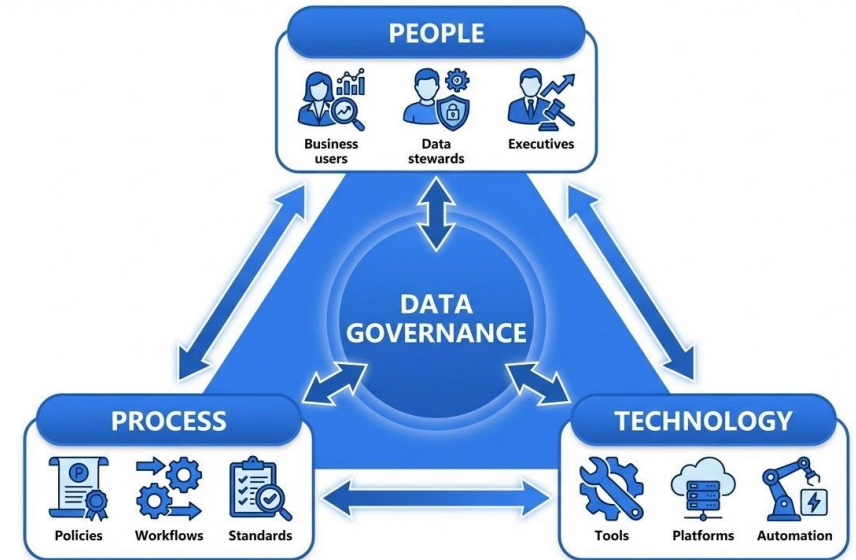
FOR BUSINESS, DATA & IT LEADERS

DEFINITION

What Is Data Governance?

Business-led orchestration of People, Process, and Technology to ensure data is trusted, secure, and usable across the enterprise.

- Formal framework for decision rights and accountability
- Business-led, not IT-led initiative
- Balances control with data democratization



Why Data Governance Matters Today

Five critical drivers making governance essential for modern enterprises



Data Volume Explosion

Exponential growth in data sources, formats, and volumes requiring structured management



Regulatory Pressure

GDPR, CCPA, HIPAA, and financial regulations demanding compliance and accountability



Cyber & Privacy Risks

Rising threats requiring robust security controls and privacy protection measures



AI/Analytics Readiness

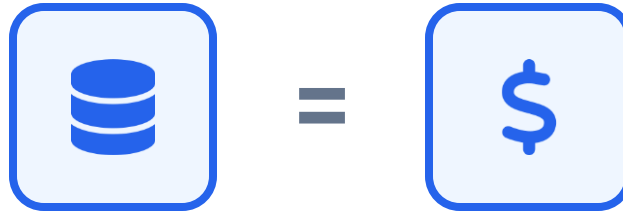
Quality and volume of data critical for AI/Analytics



Cost Control

Reducing costs of poor data management

Data as a Strategic Asset



Treat data like money: controlled, accounted for, and strategically invested. Data drives digital transformation, enhances customer experience, and powers operational excellence.



Revenue Growth

Better decisions lead to market opportunities



Customer Value

Personalized experiences drive loyalty



Operational Excellence

Streamlined processes reduce costs

Understanding Data Equity

A fundamental concept for measuring data value and risk

$$\begin{array}{c} \text{Data Equity} \\ = \\ \text{Data Assets} \\ - \\ \text{Data Liabilities} \end{array}$$



Maximize Assets



Reduce Liabilities

Data Assets vs. Data Liabilities

Understanding the Data Equity equation: maximize value, minimize risk



Data Assets

- Certified Dashboards – Trusted, governed business intelligence
- Customer 360 View – Unified, accurate customer master data
- Predictive Models – Quality data enabling AI and analytics
- Governed Datasets – Well-documented, compliant data products



Data Liabilities

- Unclassified Sensitive Data – Privacy and compliance compliance risk
- Manual Spreadsheets – Error-prone, ungoverned shadow data
- Duplicate Records – Wasted storage, incorrect decisions decisions
- Stale Reports – Outdated information leading to poor decisions

DAMA-DMBOK Framework

The Data Management Body of Knowledge (DMBOK) provides a globally recognized framework for data management and governance, establishing core governance, establishing core principles and best practices that organizations use to maximize data value.



Data is an Asset



Data Stewardship



Data Quality



Metadata Mgmt



Data Architecture



Security & Privacy



Data Lifecycle



Data Ethics



Aligning Governance with Enterprise Value

DAMA's knowledge areas create a comprehensive framework that connects data governance to business outcomes, ensuring that data management activities directly support strategic objectives, regulatory compliance, operational efficiency, and innovation.

Six Core Governance Capabilities

Essential disciplines for trusted, secure, and usable enterprise data



Metadata

Define meaning, ownership, and context through business glossaries and data catalogs



Lineage

Track data flow from source to consumption for audit and impact analysis



Data Quality

Ensure accuracy, completeness, and consistency through rules and monitoring



Architecture

Design blueprint for data systems, flows, and integration patterns



Master & Reference Data

Manage critical shared entities like customers, products, and standard codes



Operations & Monitoring

Run pipelines, track performance, and ensure data platform reliability

Metadata & Business Glossary



Definition

A Business Glossary establishes shared meaning across the organization by defining critical terms, KPIs, and data elements. It captures definitions, formulas, sources, ownership, and relationships to ensure everyone speaks the same data language.



Key Benefits



Clarity

Eliminates ambiguity by providing single source of truth for definitions



Reuse

Promotes consistent use of terms across reports, dashboards, and systems



Discoverability

Enables users to find and understand available data assets quickly

EXAMPLE TERM

Revenue

DEFINITION

Total income generated from sales of goods or services before expenses

FORMULA

`SUM(invoice_amount) WHERE status = 'completed'`

SOURCE SYSTEM

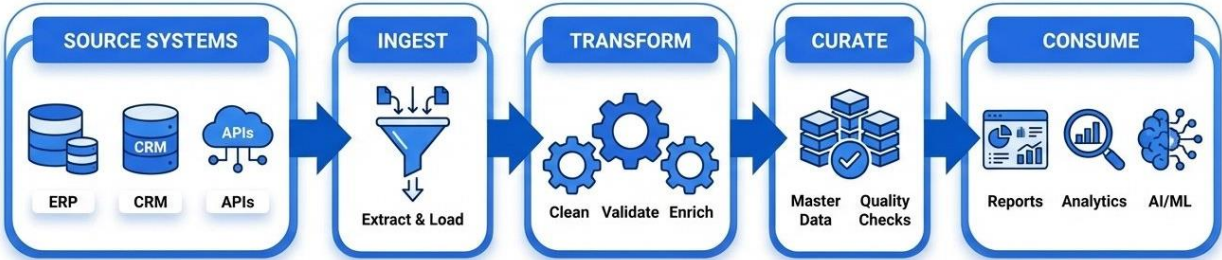
ERP.sales_transactions, CRM.orders

DATA STEWARD

Finance Team Lead (J. Smith)

Data Lineage: Source to Report

Tracing data flow from origin to consumption for transparency and control



Key Use Cases



Audit & Compliance

Prove data origin and transformations for regulatory audits



Impact Analysis

Understand downstream effects of data changes



Regulatory Reporting

Document data lineage for BCBS 239, GDPR, GDPR, SOX compliance



Incident Response

Data Quality Management

Six critical dimensions that define data quality across the enterprise



Accuracy

Data correctly represents the real-world entity or event



Completeness

All required data is present without missing values



Consistency

Data values are uniform across systems and time



Timeliness

Data is available when needed for decision-making



Validity

Data conforms to defined formats and business rules



Uniqueness

No duplicate or redundant records exist in the system



Operationalizing Quality Management

Implement quality rules, scorecards with KPIs, automated thresholds for alerting, and assign clear data owners and stewards accountable for each dimension across critical

Data Quality Dimensions

A foundational pyramid showing how quality dimensions build upon each other



Foundation Tier

Validity • Uniqueness

Base requirements ensuring data conforms to rules and has no duplicates



Middle Tier

Accuracy • Consistency

Data correctly represents reality and remains uniform across systems



Top Tier

Timeliness • Completeness

Data is available when needed with all required values present

Master Data Management (MDM)

Critical shared entities managed through golden records, match-merge, and governance workflows



Customer

Single, trusted view of customer entities across all systems and channels

- ✓ Golden Record Creation
- ✓ Identity Resolution
- ✓ Match & Merge Logic
- ✓ 360° Customer View



Product

Unified product catalog with hierarchies and attributes for consistent classification

- ✓ Product Catalog Mgmt
- ✓ Hierarchy Management
- ✓ Attribute Standardization
- ✓ SKU & Taxonomy Control



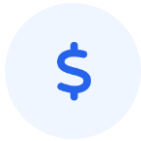
Counterparty

Manage trading partners, suppliers, and vendors with accurate relationship data

- ✓ Supplier Master Data
- ✓ Trading Partner Mgmt
- ✓ Vendor Deduplication
- ✓ Risk & Compliance Data

Reference Data

Standard code sets with versioning, approvals, and change management processes



Currency

ISO 4217 currency codes with exchange rates and regional standards

- ✓ ISO 4217 Standards
- ✓ Exchange Rate Tables
- ✓ Historical Versioning
- ✓ Regional Variants



Country

Hierarchical geographic taxonomies with ISO codes and regional groupings

- ✓ ISO 3166 Country Codes
- ✓ Geographic Taxonomy
- ✓ Region & Continent Groups
- ✓ State/Province Lists



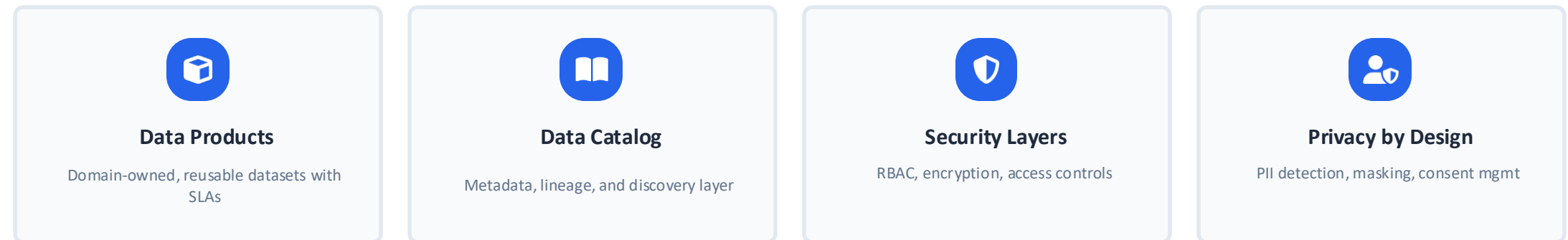
Calendar

Date hierarchies, fiscal calendars, and holiday schedules with change logs

- ✓ Fiscal Period Hierarchies
- ✓ Holiday Schedules
- ✓ Business Days Logic
- ✓ Version Control & Audit

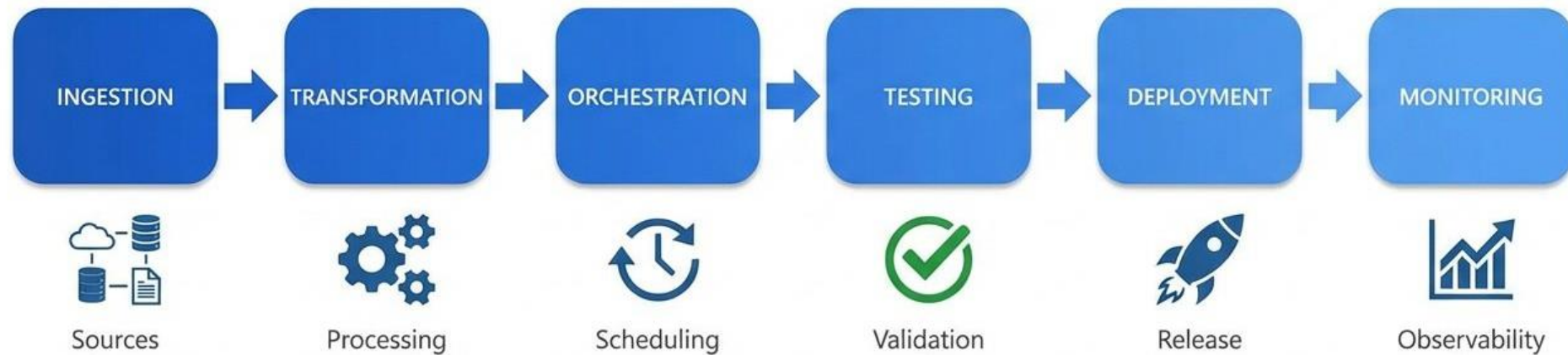
Data Architecture Blueprint

Modern lakehouse architecture with security and privacy by design



Data Operations & Pipelines

End-to-end workflow from ingestion to monitoring with operational excellence



SLAs

Service Level Agreements define contractual commitments for data delivery times, uptime guarantees, and quality thresholds that operations must meet



SLOs

Service Level Objectives are internal performance targets and KPIs that teams use to measure and optimize pipeline reliability and efficiency

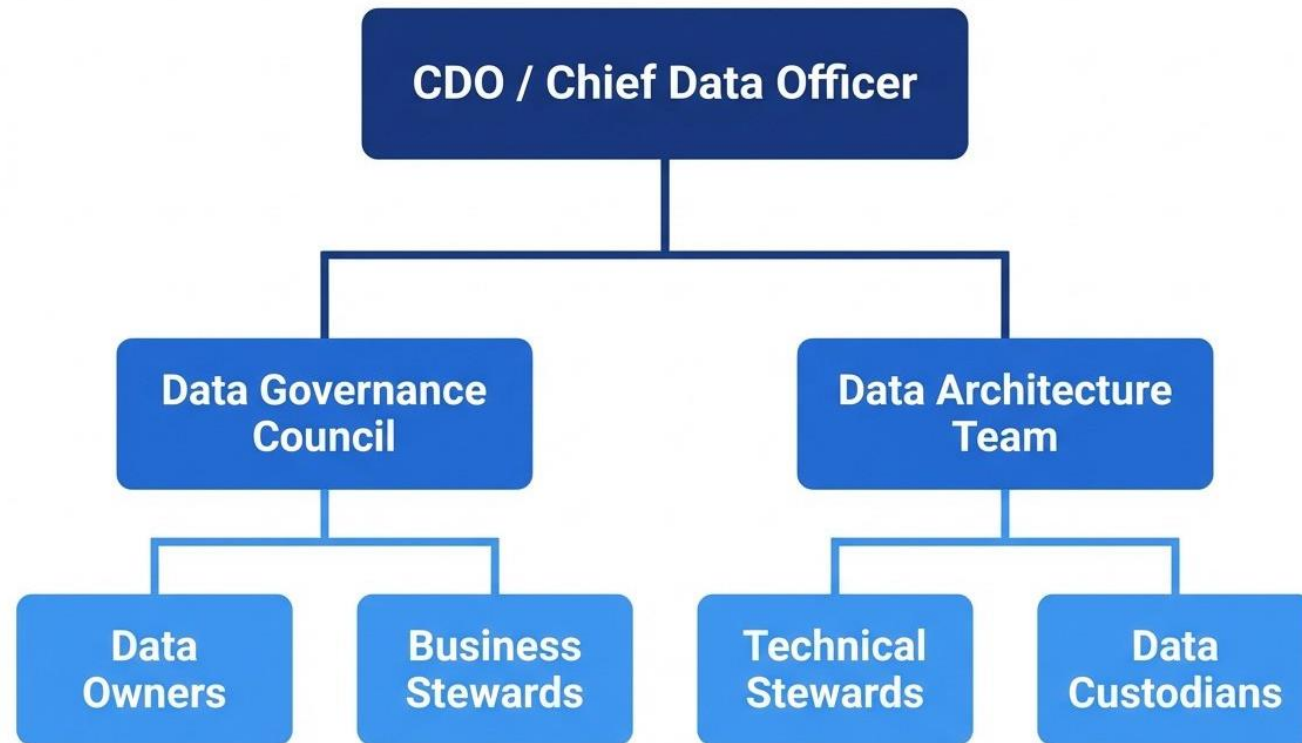


Incident Runbooks

Standardized procedures and playbooks for diagnosing, escalating, and resolving data pipeline failures and quality issues rapidly

Roles & Responsibilities

Clear accountability structure with RACI framework for decision rights



RACI Model

R

Responsible

Does the work to complete the task

A

Accountable

Ultimately answerable for correct completion

C

Consulted

Provides input and expertise

I

Informed

Kept updated on progress

Business vs. IT Responsibilities

Clear delineation of roles ensuring effective collaboration and shared accountability



Business



Define Meaning

Establish business definitions, glossary terms, and data semantics semantics



Identify Critical Data Elements

Prioritize CDEs for governance and quality management



Define Quality Rules

Specify validation rules, thresholds, and acceptance criteria



Set Policies

Create data policies, standards, and governance procedures



IT / Technical



Build Pipelines & Platform

Design and implement data infrastructure and ETL workflows



Enforce Controls

Implement security, access controls, and compliance mechanisms



Automate Quality Checks

Build automated validation, testing, and alerting systems

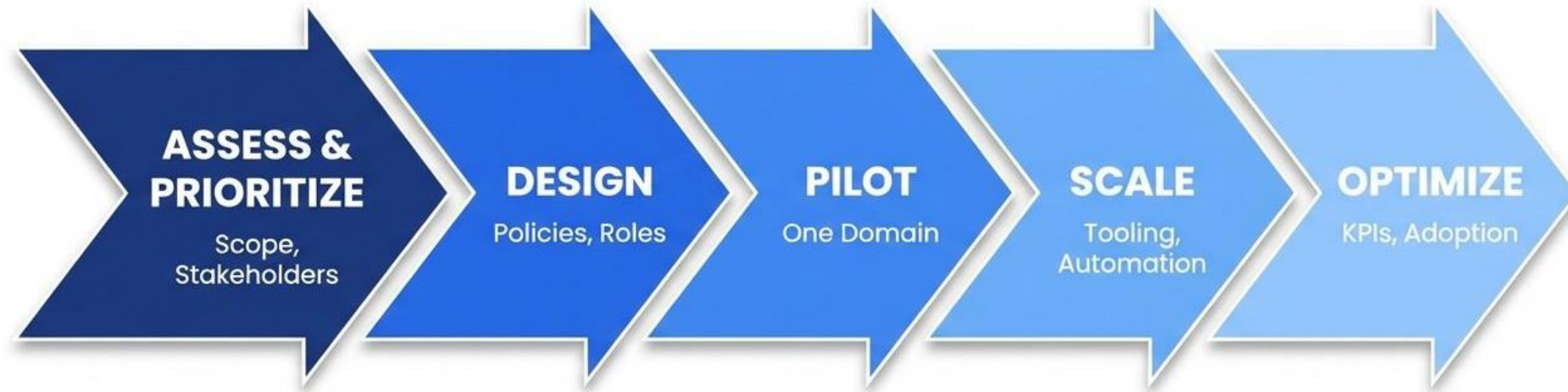


Monitor & Observe

Track data health, performance metrics, and system reliability reliability

Implementation Roadmap

Phased approach to governance with incremental value delivery and risk mitigation



1

Assess & Prioritize

Define scope, identify stakeholders, assess current state, prioritize domains

2

Design

Create policies, define roles, establish standards, design operating model

3

Pilot

Launch with one high-value domain, prove concepts, gather feedback, iterate

4

Scale

Expand to additional domains, implement tooling, automate workflows, build capabilities

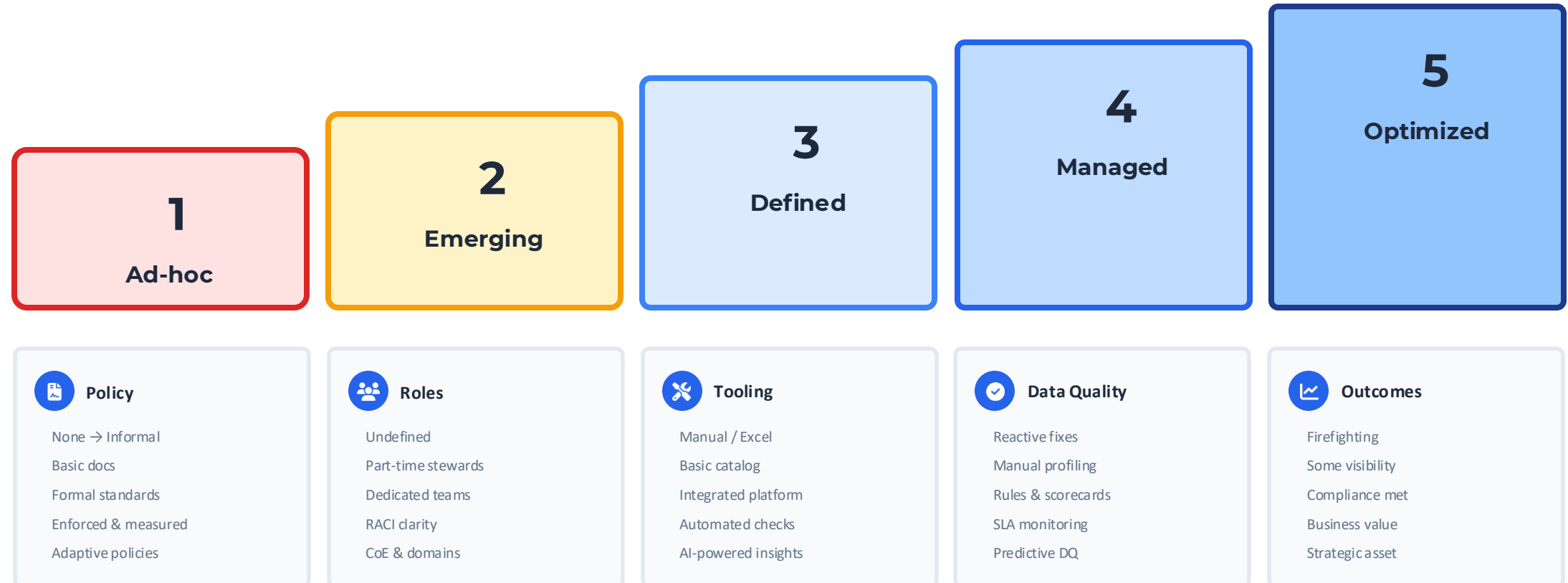
5

Optimize

Track KPIs, measure adoption, refine processes, demonstrate business value

Governance Maturity Model

Progressive evolution from ad-hoc practices to continuous optimization



Quick Wins & Best Practices

Actionable steps to launch your data governance program and demonstrate early value



Start with High-Value Domain

Choose one critical business domain (Customer, Product, Finance) where data quality issues cause pain. Demonstrate governance value quickly in this focused area.



Publish Top 50 Glossary Terms

Document and socialize the most important business terms with clear definitions, ownership, and formulas. Make them searchable and easily accessible to all users.



Define Critical Data Elements

Identify and formally designate 20-30 Critical Data Elements (CDEs) that are essential for operations, compliance, and decision-making. Prioritize governance efforts here.



Scan for PII & Sensitive Data

Run automated scans across systems to discover and classify personally identifiable information and sensitive data. Implement immediate protection measures where needed.



Automate 10 Quality Checks

Implement automated data quality rules for the most common issues: nulls, duplicates, format violations, referential integrity, and range checks. Alert data owners proactively.



Measure Adoption & Value

Track KPIs like glossary usage, data quality scores, time saved, incidents prevented, and user satisfaction. Report wins regularly to build momentum and executive support.

Automation in Data Governance

Leverage open-source and free tools to scale governance with automation



DATA CATALOG

Discovery

OpenMetadata, Amundsen

Automated metadata harvesting, search, and business glossary management

Open Source



DATA LINEAGE

Traceability

OpenLineage, Marquez

Track data flow from source to consumption with automated lineage capture

Open Source



DATA QUALITY

Validation

Great Expectations, Soda

Automated testing, profiling, and quality monitoring with assertions

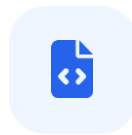
Open Source



ORCHESTRATION

Workflow

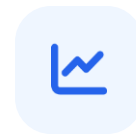
Airflow, Prefect, Dagster



POLICY AS CODE

Access Control

Open Policy Agent, Cedar



OBSERVABILITY

Monitoring

Monte Carlo, Grafana

Use Case: Banking Regulatory Reporting

Real-world implementation delivering compliance excellence and operational efficiency



Challenge

Global investment bank struggled with regulatory reporting for capital adequacy and adequacy and risk exposure, facing manual processes, data quality issues, and audit and audit failures.



Solution Approach

- ✓ Trade Data Lineage: End-to-end traceability from trade capture to risk reports
- ✓ Critical Data Elements: Defined CDEs for capital ratios, counterparty risk, and exposure metrics
- ✓ Automated Quality Checks: Real-time validation rules with 24/7 monitoring and alerting
- ✓ Audit Trails: Complete change history, approval workflows, and regulatory regulatory evidence



60% Faster

Regulatory Compliance



85% Reduction

Data Quality Incidents



100% Confidence

Regulatory Audits Passed

Key Takeaways

Essential principles for building a successful data governance program

1

Governance builds trust — and trust enables analytics, AI, and data-driven decisions



2

Combine clear roles, a pragmatic roadmap, and automation to scale value safely



3

Start small with quick wins, then expand systematically across the enterprise



Ready to Begin Your Governance Journey?

Apply these principles to transform your data into a trusted, strategic asset that powers business outcomes

Get Started →